KAIZEN MATHS

Lesson starter

Fraction Arithmetic and Mixed Numbers

All Operations

Name _____

S1

$\frac{2}{4} + \frac{3}{5}$

S2

$\frac{4}{5} - \frac{1}{7}$

S3

$\frac{3}{8} \times \frac{3}{6}$

S4

$\frac{1}{4} \div \frac{8}{9}$

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NUM-4395-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Fraction Arithmetic and Mixed Numbers

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E1

$\frac{3}{5} \times (\frac{4}{7} + 1\frac{3}{5}) \div \frac{8}{9} + \frac{3}{5} - \frac{1}{5}$

E2

$\frac{5}{6} + \frac{7}{9}$

E3

$\frac{3}{4} - \frac{4}{6}$

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NUM-4395-L4 · E

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$\frac{3}{4} - \frac{4}{6}$

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Show your working

Teacher answers & working

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S1

$\frac{2}{4} + \frac{3}{5}$

Answer:

$1\frac{1}{10}$

Working:

Use the cross-multiplication denominator method.

$\frac{2}{4} + \frac{3}{5} = \frac{(2\times5)+(3\times4)}{4\times5}$

$\frac{(10)+(12)}{20} = \frac{22}{20}$

The highest common factor of 22 and 20 is 2.

$\frac{22}{20} = \frac{22\div2}{20\div2} = 1\frac{1}{10}$

Final answer: $1\frac{1}{10}$

S2

$\frac{4}{5} - \frac{1}{7}$

Answer:

$\frac{23}{35}$

Working:

Use the cross-multiplication denominator method.

$\frac{4}{5} - \frac{1}{7} = \frac{(4\times7)-(1\times5)}{5\times7}$

$\frac{(28)-(5)}{35} = \frac{23}{35}$

$\frac{23}{35}$ is already in simplest form.

Final answer: $\frac{23}{35}$

S3

$\frac{3}{8} \times \frac{3}{6}$

Answer:

$\frac{3}{16}$

Working:

Multiply the numerators and multiply the denominators.

$\frac{3}{8} \times \frac{3}{6} = \frac{3\times3}{8\times6}$

$\frac{3\times3}{8\times6} = \frac{9}{48}$

The highest common factor of 9 and 48 is 3.

$\frac{9}{48} = \frac{9\div3}{48\div3} = \frac{3}{16}$

Final answer: $\frac{3}{16}$

Teacher answers & working

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S4

$\frac{1}{4} \div \frac{8}{9}$

Answer:

$\frac{9}{32}$

Working:

Keep the first fraction and multiply by the reciprocal of the second fraction.

$\frac{1}{4} \div \frac{8}{9} = \frac{1}{4} \times \frac{9}{8}$

$\frac{1}{4} \times \frac{9}{8} = \frac{1 \times 9}{4 \times 8}$

$\frac{1 \times 9}{4 \times 8} = \frac{9}{32}$

$\frac{9}{32}$ is already in simplest form.

Final answer: $\frac{9}{32}$

Teacher answers & working

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E1

$\frac{3}{5} \times (\frac{4}{7} + 1\frac{3}{5}) \div \frac{8}{9} + \frac{3}{5} - \frac{1}{5}$

Answer:

$1\frac{303}{350}$

Working:

Work inside the brackets first.

$1\frac{3}{5} = \frac{1\times5+3}{5} = \frac{8}{5}$

Use the cross-multiplication denominator method.

$\frac{4}{7} + \frac{8}{5} = \frac{(4\times5)+(8\times7)}{7\times5}$

$\frac{(20)+(56)}{35} = \frac{76}{35}$

$\frac{76}{35}$ is already in simplest form.

The bracket is $2\frac{6}{35}$.

Now multiply by the fraction before the bracket.

Multiply the numerators and multiply the denominators.

$\frac{3}{5} \times \frac{76}{35} = \frac{3\times76}{5\times35}$

$\frac{3\times76}{5\times35} = \frac{228}{175}$

$\frac{228}{175}$ is already in simplest form.

After multiplication, the expression is $1\frac{53}{175} \div \frac{8}{9} + \frac{3}{5} - \frac{1}{5}$.

Now divide by the next fraction.

Keep the first fraction and multiply by the reciprocal of the second fraction.

$\frac{228}{175} \div \frac{8}{9} = \frac{228}{175} \times \frac{9}{8}$

$\frac{228}{175} \times \frac{9}{8} = \frac{228\times9}{175\times8}$

$\frac{228\times9}{175\times8} = \frac{2052}{1400}$

The highest common factor of 2052 and 1400 is 4.

$\frac{2052}{1400} = \frac{2052\div4}{1400\div4} = 1\frac{163}{350}$

After division, the expression is $1\frac{163}{350} + \frac{3}{5} - \frac{1}{5}$.

Now add the next fraction.

Use the cross-multiplication denominator method.

$\frac{513}{350} + \frac{3}{5} = \frac{(513\times5)+(3\times350)}{350\times5}$

$\frac{(2565)+(1050)}{1750} = \frac{3615}{1750}$

The highest common factor of 3615 and 1750 is 5.

$\frac{3615}{1750} = \frac{3615\div5}{1750\div5} = 2\frac{23}{350}$

After addition, the expression is $2\frac{23}{350} - \frac{1}{5}$.

Now subtract the final fraction.

Use the cross-multiplication denominator method.

Teacher answers & working

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E1 (continued)

$\frac{723}{350} - \frac{1}{5} = \frac{(723\times5)-(1\times350)}{350\times5}$

$\frac{(3615)-(350)}{1750} = \frac{3265}{1750}$

The highest common factor of 3265 and 1750 is 5.

$\frac{3265}{1750} = \frac{3265\div5}{1750\div5} = 1\frac{303}{350}$

Final answer: $1\frac{303}{350}$

E2

$\frac{5}{6} + \frac{7}{9}$

Answer:

$1\frac{11}{18}$

Working:

Use the cross-multiplication denominator method.

$\frac{5}{6} + \frac{7}{9} = \frac{(5\times9)+(7\times6)}{6\times9}$

$\frac{(45)+(42)}{54} = \frac{87}{54}$

The highest common factor of 87 and 54 is 3.

$\frac{87}{54} = \frac{87\div3}{54\div3} = 1\frac{11}{18}$

Final answer: $1\frac{11}{18}$

E3

$\frac{3}{4}$ - $\frac{4}{6}$

Answer:

$\frac{1}{12}$

Working:

Use the cross-multiplication denominator method.

$\frac{3}{4} - \frac{4}{6} = \frac{(3\times6)-(4\times4)}{4\times6}$

$\frac{(18)-(16)}{24} = \frac{2}{24}$

The highest common factor of 2 and 24 is 2.

$\frac{2}{24} = \frac{2\div2}{24\div2} = \frac{1}{12}$

Final answer: $\frac{1}{12}$